Practice Problems 5

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 4.6

2. Given $A = \begin{bmatrix} 0 & 4 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -8 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 & 9 \\ 6 & 1 & 1 \end{bmatrix}$, verify that

   (a) $(A + B)' = A' + B'$

   (b) $(AC)' = C'A'$

6. Let $A = I - X\,(X'X)^{-1}\,X'$.

   (a) Must A be square? Must $(X'X)$ be square? Must X be square?

   (b) Show that matrix A is idempotent.